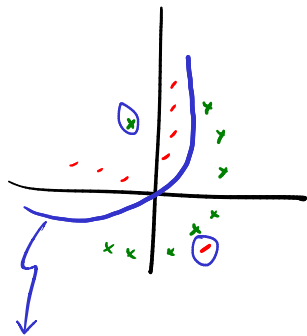


WEEK-XI

SOFT MARGIN
PRIMAL SVM



margin $\leftarrow$ length(w)

min $w \in \mathbb{R}^d$, ξ (primal variables)

$$\frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i$$

Hyperparameter (≥ 0)

St constraint 1: $(w^T x_i) y_i + \xi_i \geq 1$ $+i$

(α) constraint 2: $\xi_i \geq 0$ $+i$

$\rightarrow 1 - w^T x_i y_i - \xi_i \leq 0$

$\rightarrow -\xi_i \leq 0$

x_i (green letter) (bribes slack variables)

Quadratically separable

Lagrangian f^n

$$L(w, \xi, \alpha, \beta) = \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i + \sum_{i=1}^n \alpha_i (1 - w^T x_i y_i - \xi_i)$$

$d_i(w) \leq 0$

$$+ \sum_{i=1}^n \beta_i (-\xi_i)$$

DUAL PROBLEM

Min-Max problem $\rightarrow$ min w, ξ $\left[\max_{\alpha \geq 0, \beta \geq 0} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i + \sum_{i=1}^n \alpha_i (1 - w^T x_i y_i - \xi_i) + \sum_{i=1}^n \beta_i (-\xi_i) \right]$

Objective & constraints are convex f^n ||| STRONG DUALITY

Max-Min problem $\rightarrow$ max $\alpha \geq 0, \beta \geq 0$ $\left[\min_{w, \xi} \left[\frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i + \sum_{i=1}^n \alpha_i (1 - w^T x_i y_i - \xi_i) + \sum_{i=1}^n \beta_i (-\xi_i) \right] \right]$

L

• Fix α, β

• $\frac{\partial L}{\partial w} = 0 \Rightarrow \boxed{w_{\alpha, \beta}^* = \sum_{i=1}^n \alpha_i x_i y_i} - ①$ $\left| \begin{matrix} w_{\alpha, \beta}^* \\ x y \alpha \end{matrix} \right.$

• $\frac{\partial L}{\partial \xi_i} = 0 \Rightarrow C + \alpha_i (-1) + \beta_i (-1) = 0$

$\Rightarrow \boxed{\alpha_i + \beta_i = C} - ②$

Back Substituting $w_{\alpha, \beta}^*$ into the Lagrangian

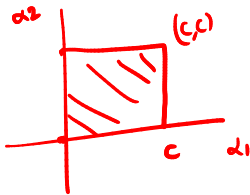
Dual problem

- β is restricting range of α

$$\begin{aligned} \max \quad & \alpha^T 1 - \frac{1}{2} \alpha^T Y^T X^T X Y \alpha \\ \alpha \geq 0 \\ \beta \geq 0 \\ \alpha + \beta = C \end{aligned}$$

no β term

III



$$\max \quad \alpha^T 1 - \frac{1}{2} \alpha^T Y^T X^T X Y \alpha$$

$0 \leq \alpha \leq C$

Kernelizable

Hyperparameter

Box Constraints

if

$$C = 0$$

$\Rightarrow$

$$\alpha^* = 0 \in \mathbb{R}^n$$

$$\Rightarrow w^* = \sum \alpha_i x_i y_i$$

$$\Rightarrow \boxed{w^* = 0}$$

$$C = \infty \Rightarrow \text{Hard-margin}$$

What do the COMPLEMENTARY SLACKNESS (C-S)
Say about the Soft-margin SVM?

Let (w^*, ξ^*) be the primal optimal solutions.

Let (α^*, β^*) be the dual optimal solutions.

C-S

①

$$\forall i \quad \alpha_i^* (1 - w^{*T} x_i y_i - \xi_i^*) = 0$$

②

$$\forall i \quad \beta_i^* (\xi_i^*) = 0$$

- Complementary Slackness says, "The extra term(s) we add to the Lagrangian would be **zero** at optimality".

Various cases possible

①

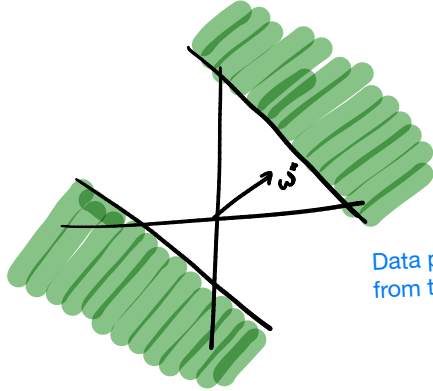
$$\alpha_i^* = 0$$

$\Rightarrow$

$$\beta_i^* = C \quad [\alpha_i^* + \beta_i^* = C]$$

$\Downarrow$ C-S ②

$$\xi_i^* = 0$$



we know $w^T x_i y_i + \xi_i^* \geq 1$ (Primal feasibility)

$\Rightarrow w^T x_i y_i \geq 1 \Rightarrow w^*$ classifies (x_i, y_i) correctly!

$\alpha_i^* = 0$
 $\Downarrow$

Data points could be on or away from the supporting hyperplane. $\Rightarrow$ not contributing in finding w^*

②

$\alpha_i^* \in (0, c)$

$0 < \alpha_i^* < c$

$\Rightarrow [c \cdot s \text{ ①}]$

strictly positive quantity $\Rightarrow$

$\beta_i^* \in (0, c)$

$0 < \beta_i^* < c$

$1 - w^T x_i y_i - \xi_i^* = 0$

$\Downarrow$
 $1 - w^T x_i y_i = 0$

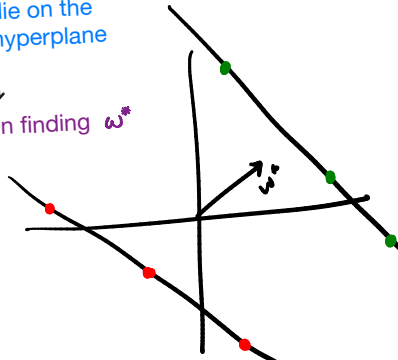
$\Rightarrow \xi_i^* = 0$ [c-s ②]

$\Rightarrow \boxed{w^T x_i y_i = 1}$

$0 < \alpha_i^* < c$

Data points lie on the supporting hyperplane

$\Downarrow$
 contributing in finding w^*



③

$\alpha_i^* = c$

$\Rightarrow [c \cdot s \text{ ①}]$

$1 - w^T x_i y_i - \xi_i^* = 0$

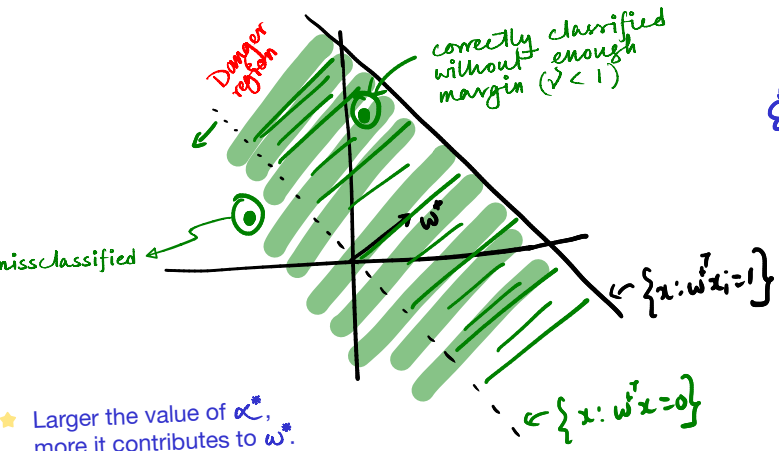
$\Downarrow$
 $\beta_i^* = 0$

$\Downarrow$
 $\xi_i^* \geq 0$

$\xi_i^* = 1 - w^T x_i y_i \geq 0$

$\Rightarrow \boxed{w^T x_i y_i \leq 1}$

• Support vectors: any vector for which $\alpha_i^* > 0$.



★ Larger the value of α_i^* , more it contributes to w^* .

$\alpha_i^* = c$ contributes most.

Points where either x_i is incorrectly classified by w^* or correctly classified but with margin ≤ 1

Let's see this from the primal point of view.

Case ①

$\boxed{w^T x_i y_i < 1} \Rightarrow$ Margin violation (margin < 1)

$w^T x_i y_i + \xi_i^* \geq 1 \Rightarrow$ (Primal feasibility)

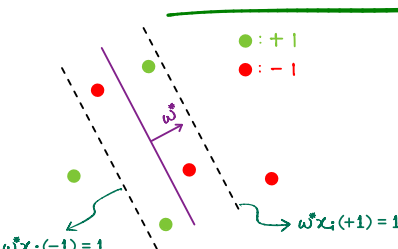
$\xi_i^* \geq 1 - \frac{w^T x_i y_i}{< 1}$

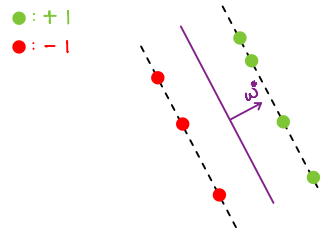
$\Rightarrow \xi_i^* > 0 \xRightarrow{c \cdot s \text{ ②}} \beta_i^* = 0 \Rightarrow \boxed{\alpha_i^* = c}$

$w^T x_i y_i < 1$ and $\alpha_i^* = c$

Point is strictly away from the hyperplane, but on the wrong side.

Point is very important as $\alpha_i^* = c$ for the point.





Case ②

(margin = 1)
 $\omega^T x_i y_i = 1 \Rightarrow$ Data points are on the supporting hyperplane

$\xi_i^* \geq 1 - \omega^T x_i y_i$ (Primal feasibility)

$\xi_i^* \geq 0 \Rightarrow \alpha_i^* \in [0, C]$

$\beta_i^* \in [0, C]$

Since the objective penalises $\xi_i \xi_i^*$, the optimiser will choose the smallest feasible value, i.e., $\xi_i^* = 0$.

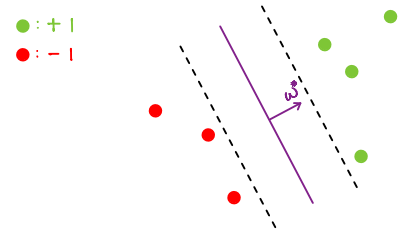
Points exactly on the margin are correctly classified and have zero slack.

Case ③

$\omega^T x_i y_i > 1$
 (margin > 1) $\Rightarrow$ Data points are well away from the supporting hyperplane

$\Rightarrow \underbrace{1 - \omega^T x_i y_i}_{< 0} - \underbrace{\xi_i}_{\geq 0} < 0$

as ① $\Rightarrow \alpha_i^* = 0$



Points are not important as they do not contribute to ω^* .

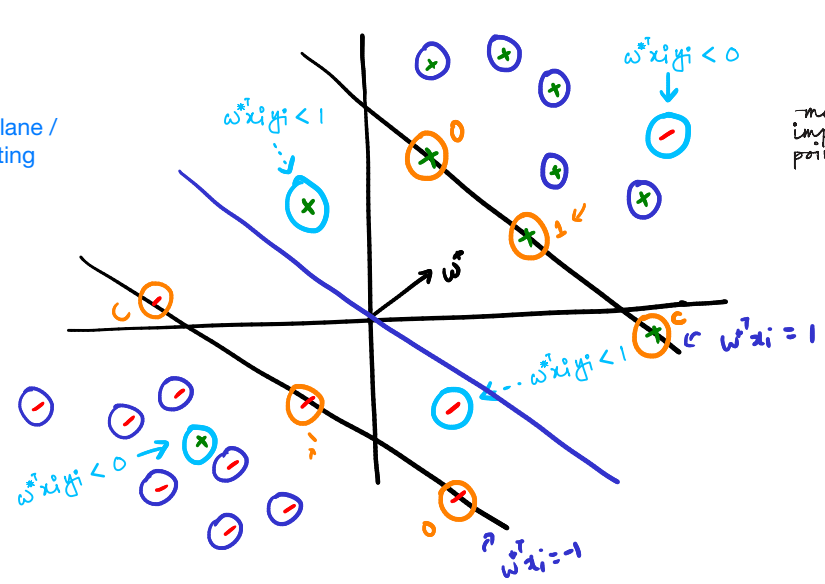
SUMMARY

Dual POV	$\alpha_i^* = 0$	$\Rightarrow$	$\omega^T x_i y_i \geq 1 \rightarrow$ Data points are on or away from the supporting hyperplane.
	$0 < \alpha_i^* < C$	$\Rightarrow$	$\omega^T x_i y_i = 1 \rightarrow$ Data points are on the supporting hyperplane.
	$\alpha_i^* = C$	$\Rightarrow$	$\omega^T x_i y_i \leq 1 \rightarrow$ Data points are inside the margin or on the supporting hyperplane or misclassified.

Primal POV	$\omega^T x_i y_i < 1$	$\Rightarrow$	$\alpha_i^* = C$
	$\omega^T x_i y_i = 1$	$\Rightarrow$	$0 \leq \alpha_i^* \leq C$
	$\omega^T x_i y_i > 1$	$\Rightarrow$	$\alpha_i^* = 0$

* Points on the supporting hyperplane / on the wrong side of the supporting hyperplane contribute to ω^* .

"Sparse solution"



$\gamma > 1 \rightarrow \alpha_i = 0$
 $\gamma = 1 \rightarrow \alpha_i = C$
 $\gamma < 1 \rightarrow \alpha_i \in [0, C]$

most important points $\rightarrow$

So far....

Generative

- Naive - Bayes
- G-D-A / G-N-B

Discriminative

- K-NN
- Decision trees
- Logistic regression
- Perceptron
- Support-vector machines

- **Ensemble:** a predictive model that combines the outputs of multiple individual models (base learners) to produce a single, final prediction.

Goal :- Meta Classifiers (or) Ensemble Classifiers.

WEAK LEARNERS $\longrightarrow$ STRONG LEARNERS.

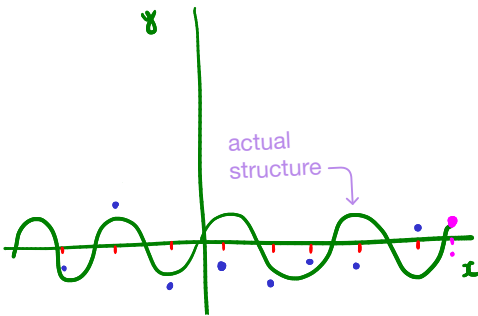
(better than random) $\longrightarrow$ 50% accuracy

Overfitting

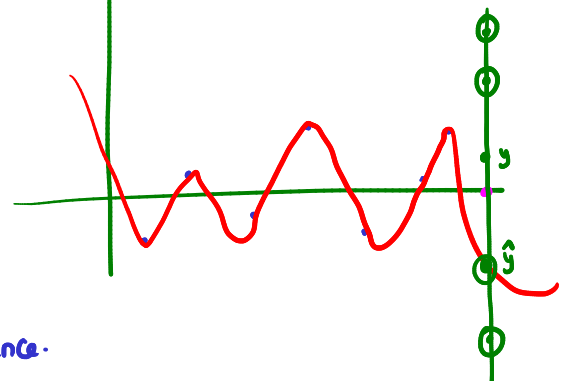
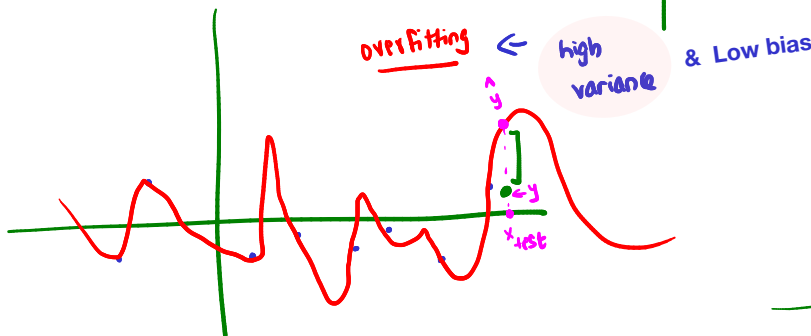
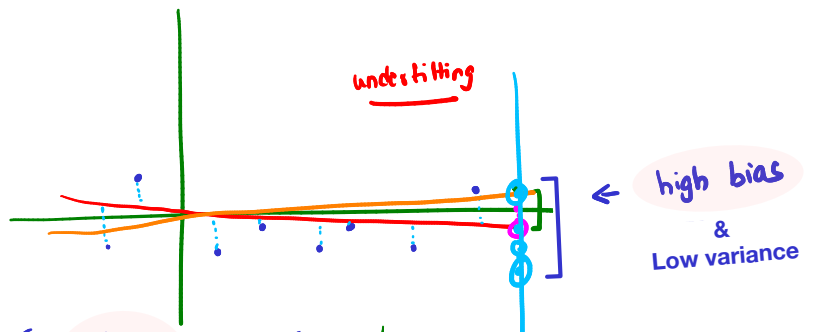
- fitting noise thinking it is part of the structure.

Underfitting

- missing out on structure thinking it is noise.



- **Bias:** how good the assumed structure is representing the truth.
- **Variance:** how much variation is observed in the assumed structure to fluctuations in the training dataset.



Though it captures the core idea, but not a formal mathematical definition of **Bias-Variance Tradeoff**

$$\text{Error} = \text{bias} + \text{variance}.$$

BAGGING - BOOTSTRAP AGGREGATING

$$\{x_1, x_2, \dots, x_n\} \sim \mathcal{N}(\mu, 1)$$

known variance

$$\begin{aligned} \hat{\mu}_1 &= x_1 \\ \hat{\mu}_2 &= x_2 \\ &\vdots \\ \hat{\mu}_n &= x_n \end{aligned}$$

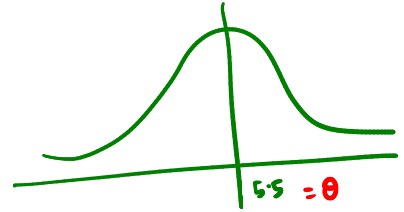
unknown mean

$$\hat{\mu}_{ML} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$E[\hat{\mu}_{ML}] = \mu$$

$$E[\hat{\theta}] = \theta \leftarrow \text{unbiased estimators}$$

$$E[\hat{\mu}_i] = E[x_i] = \mu$$



$$\{5, 7.3, 6.7, 8.2, \dots, 9\}$$

$$\hat{\mu}_1 = 5$$

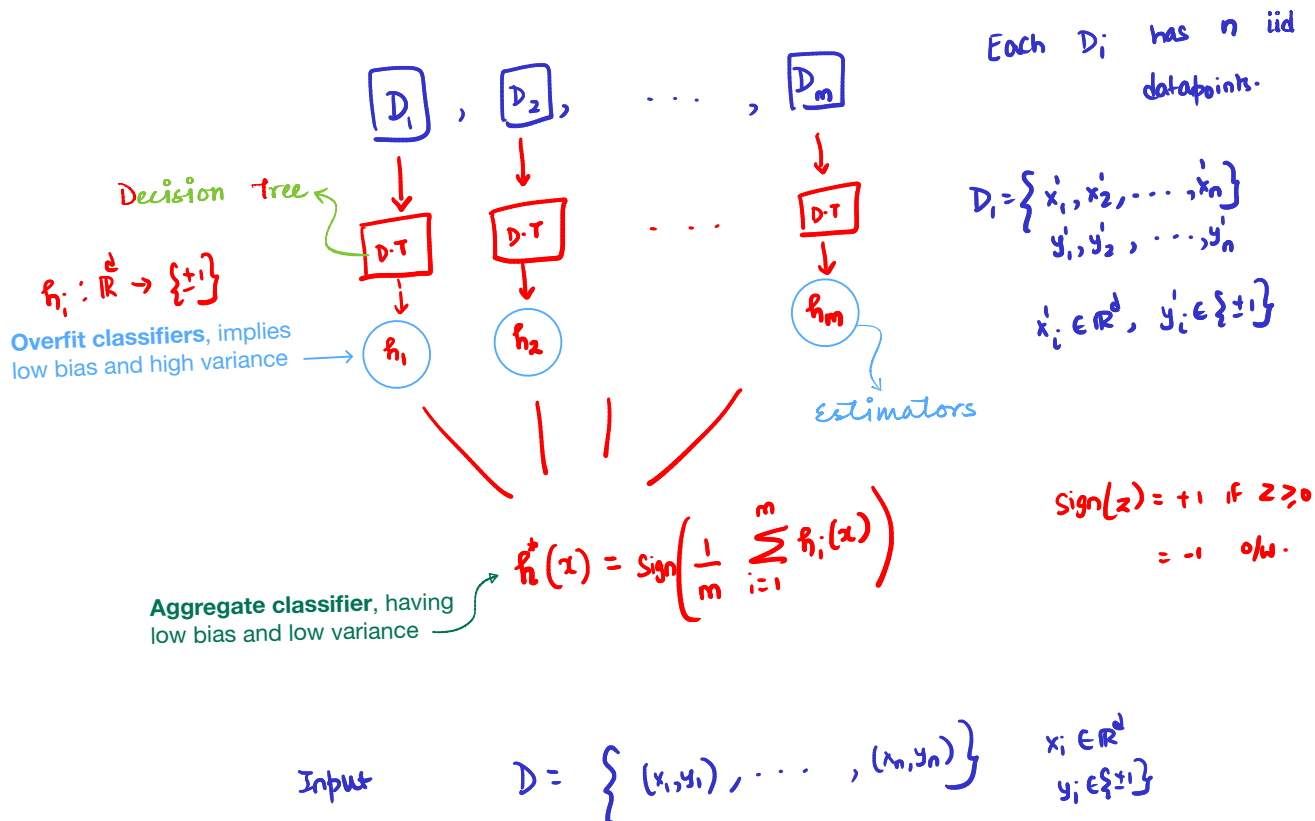
$$\hat{\mu}_2 = 7.3$$

$$\vdots$$

$$\hat{\mu}_n = 9$$

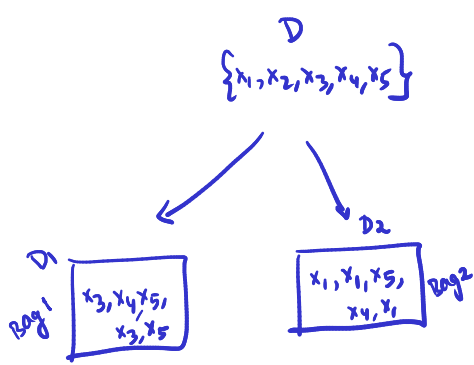
"Averaging reduces variance" → when the samples are iid.

Bagging



1. Create m different "bags" (datasets) using the given dataset by uniformly sampling with replacement.
2. Train m estimators on m different bags.
3. Aggregate the outputs of all m estimators.

Bootstrapping → uniform sampling with replacement



Probability of a point missing from a bag →

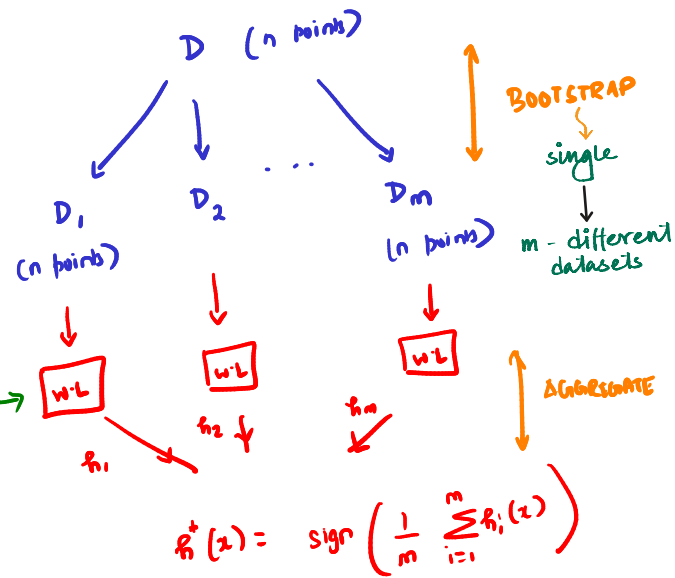
$$P(x_j \in D_i) = 1 - \left(1 - \frac{1}{n}\right)^n \approx 1 - \frac{1}{e} \approx 63\%$$

≈ 37% for loge n

Probability of x_j belongs to D_i

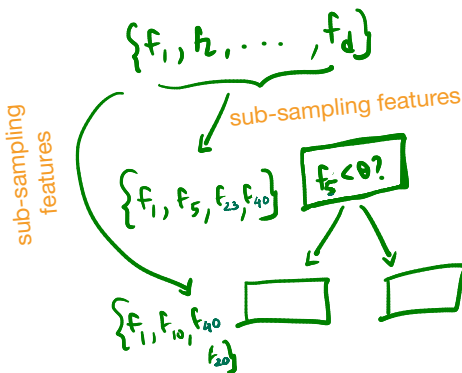
Adding more weak learners result in reduction in variance, but the bias stays almost the same.

⇒ * Bagging reduces variance!



If the weak learners are de-correlated, then the reduction in variance would be large.

Random forest: feature-bagged decision trees



- Bags Decision trees (typically over fit trees) [training error = 0] (low bias)

- Feature bagging ($\sqrt{d}$)
 - Trees get decorrelated across bags
 - # features bagged in classification
 - "Random Subspace method"
 - Ensemble learning technique (Extension of Bagging)

1. We have dataset with d features.
2. For each base learner:
 - Select a subset of features
 - Train the learner using only those features
3. Combine predictions from all learners.

- Bootstrapping - uniform Sampling with replacement
- Bag - Averaging

BOOSTING

BOOSTING [ADABOOST]

L FREUND & SCHAPIRE (Gödel Prize)

- In BOOSTING, these weak learners need not be overfit classifiers, these could be underfit classifiers as well.

Weak Learner

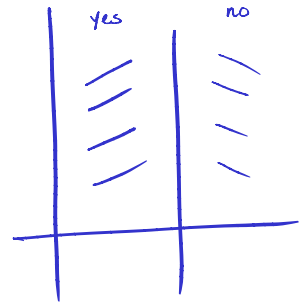
Strong Learner

Decision Stumps

$$f_x < \theta$$

+1 -1

1-level or 2-level



- Decision Stump** is a Decision Tree, which uses only a single attribute for splitting. For discrete attributes, this typically means that the tree consists only of a single interior node (i.e., the root has only leaves as successor nodes). If the attribute is numerical, the tree may be more complex.

ADA- BOOST ALGORITHM

↳ Adaptive.

Input: $S = \{ (x_1, y_1), \dots, (x_n, y_n) \}$

$x_i \in \mathbb{R}^d$
 $y_i \in \{ \pm 1 \}$

Uniform sampling with replacement

Initialize

$D_0(i) = \frac{1}{n}$: At initialisation, every datapoint, i gets an uniform weight.

iteration

for $t = 1, \dots, T$

(distribution)

weights associated with datapoints

Input (S, D_{t-1}) to a weak learner to get h_t

Higher weight implies to higher probability of getting chosen, means datapoints with higher weight are repeated more in the dataset.

$h_t =$

$$h_t: \mathbb{R}^d \rightarrow \{ \pm 1 \}$$

$$\tilde{D}_t(i) = \begin{cases} D_{t-1}(i) \cdot e^{\alpha_t} & \text{if } h_t(x_i) \neq y_i \\ D_{t-1}(i) \cdot e^{-\alpha_t} & \text{if } h_t(x_i) = y_i \end{cases}$$

misclassified
correctly classified

$$D_{t+1}(i) = \frac{\tilde{D}_t(i)}{\sum_j \tilde{D}_t(j)} : \text{normalising}$$

End

- If the Weak Learner is unable to handle weights, we can create a bag from the dataset according to the weights associated with datapoints.

- Weighted Sampling** - misclassified points by the current weak learner get higher weightage for being selected in the training set for the next weak learner

$$D_0 = \begin{bmatrix} 0.33 & 0.33 & 0.33 \end{bmatrix} \leftarrow$$

$$D_1 = \begin{bmatrix} 0.3 & 0.5 & 0.2 \end{bmatrix} \leftarrow h_1$$

$$D_2 = \begin{bmatrix} 0.3 \cdot e^{\alpha_1} & 0.5 \cdot e^{-\alpha_1} & 0.2 \cdot e^{\alpha_1} \end{bmatrix}$$

$$Z = 0.3 e^{\alpha_1} + 0.5 e^{-\alpha_1} + 0.2 e^{\alpha_1}$$

$$\text{error}(h_t) = \sum_{h_t(x_i) \neq y_i} D_{t-1}(i)$$

$h_1, h_2, \dots, h_T$

$$h_T^*(x) = \text{sign} \left(\sum_{t=1}^T \alpha_t h_t(x) \right) \quad \alpha_t = \ln \sqrt{\frac{1 - \text{error}(h_t)}{\text{error}(h_t)}}$$

one can prove

If $T \geq \frac{1}{2\gamma^2} \ln(2n)$, then

• $T \propto \frac{n}{\gamma^2}$

Training error = 0

How much more the weak learner is better than the random!

How good is my weak learner?

In practice,



• Can't run in parallel
→ as it uses iteration.

- * After the training error reaches zero, boosting effectively continues to refine the classifier. As additional weak learners are added, they reshape the decision boundary in a way that increases the margin of the combined classifier. A larger margin is associated with better generalisation, meaning improved performance on unseen test data. As a result, boosting can continue to improve test accuracy even after the training error has already dropped to zero.